

Program: Diploma in Automation and Robotics	
Course Code: 5333A	Course Title: Industrial Robotics
Semester: 5	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To familiarize automation and brief history of robot and applications.
- To study the control strategies used in industrial robots.
- To give knowledge about robot end effectors and their design.
- To learn about work cell design and robot cycle time analysis.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Knowledge in Robotics		Foundations of Robotics , Robotic Case Studies	2,3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Understand different configurations of robots and different control strategies and controllers	15	Understanding
CO2	Understand about different type of actuators and sensor devices used in robots.	15	Understanding
CO3	Understand Material Transfer and handling applications of Robots	14	Understanding
CO4	Understand about work cell design and robot cycle time analysis.	14	Understanding
	Series Test	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand different configurations of robots and different control strategies and controllers		
M1.01	Understand Automation and Robot anatomy	3	Understanding
M1.02	Understand different configuration of robots and joint notation schemes.	4	Understanding
M1.03	Understand Basic control system models.	4	Understanding
M1.04	Understand controllers used in Robot system models.	4	Understanding
Contents: Automation and Robotics, Robot anatomy, configuration of robots, joint notation schemes, work volume, Basic control system models, slew motion, joint – interpolated motion and straight-line motion, controllers like on/off, proportional, integral, proportional plus integral, proportional plus derivative, proportional plus integral plus derivative.			
CO2	Understand about different type of actuators and sensor devices used in robots		
M2.01	Understand about actuation and feedback components.	3	Understanding
M2.02	Understand about position / velocity sensors, actuators and power transmission devices.	4	Understanding
M2.03	Understand about different type of Grippers.	4	Understanding
M2.04	Understand different methods of power and control signals to end effectors.	4	Understanding

	Series Test – I	1	
Contents: Robot actuation and feedback components position and velocity sensors, actuators and power transmission devices, mechanical grippers, vacuum cups, magnetic grippers, pneumatic, electric, hydraulic and mechanical methods of power and control signals to end effectors.			
CO3	Understand Material Transfer and handling applications of Robots		
M3.01	Understand about the general considerations in robot material handling and transfer applications.	4	Understanding
M3.02	Understand about pick and place operations and palletizing operations	3	Understanding
M3.03	Understand about machine loading and unloading operations	3	Understanding
M3.04	Understand about molding, forging, machining and stamping operations using robots	4	Understanding
Contents: General considerations in robot material handling, material transfer applications, pick and place operations, palletizing and related operations, machine loading and unloading, die casting, plastic molding, forging, machining operations, stamping press operations using robots.			
CO4	Understand about work cell design and robot cycle time analysis.		
M4.01	Understand about Robot cell layouts and Robots - machine interface.	4	Understanding
M4.02	Understand about work cell design and work cell control	3	Understanding
M4.03	Understand about error detection and recovery in robotic systems	3	Understanding
M4.04	Understand about robot cycle time analysis.	4	Understanding
	Series Test – II	1	
Contents: Robot cell layouts, multiple Robots and machine interface, other considerations in work cell design, work cell control, interlocks, error detection and recovery, work cell controller, robot cycle time analysis.			

Text / Reference

T/R	Book Title/Author
T1	Richard D. Klafter, Thomas A. Chmielewski and Michael Negin, "Robotic Engineering - An Integrated Approach", Prentice Hall India, 2002
R1	Deb S.R., " Robotics Technology and Flexible Automation ", Tata McGraw-Hill, Publishing Co., Ltd., 1994.
R2	K.S. Fu., R.C.Gonzalez, C.S.G.Lee, " Robotics Control Sensing ", Vision and Intelligence, McGraw Hill International Edition, 1987.
R3	Mikell P. Groover, Mitchell Weiss, "Industrial Robotics, Technology, Programming and Applications ", McGraw Hill International Editions, 1st Edition, 2000